

AS1109 Student Handout

Chain Rule, Dependency Diagrams, and Implicit Differentiation

Based on this week's Calculus & ODE classes

What this handout is for

This handout is meant to help you understand three connected ideas in a simple, step-by-step way:

1. how the **multivariable chain rule** works,
2. how to read a **dependency diagram** (also called a dependency tree/graph), and
3. why **implicit differentiation** is really just the **chain rule applied to an equation that stays equal to 0**.

Prerequisites

You only need:

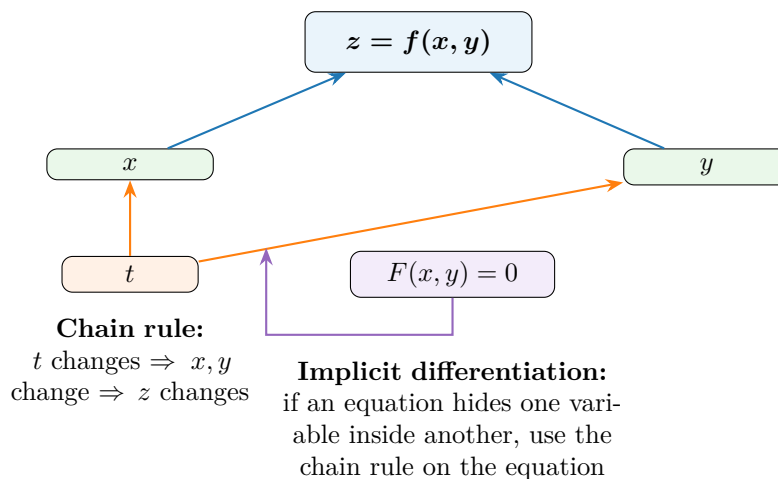
- XII-standard differentiation,
- the idea of a **partial derivative**,
- the total differential

$$df = f_x dx + f_y dy.$$

1 The big picture

The central idea

If $z = f(x, y)$, then z depends on x and y . But if x and y themselves depend on some other variable(s), then z also changes because of those hidden dependencies.



2 One-parameter chain rule: $z = f(x, y)$, $x = x(t)$, $y = y(t)$

Step 1: Start from the total differential

From class, we know:

$$dz = f_x dx + f_y dy.$$

If x and y both depend on t , then divide by dt :

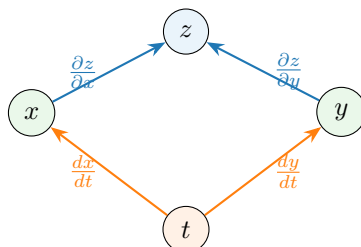
$$\frac{dz}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt}$$

or equivalently,

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}.$$

Memory rule

Multiply along each path, then add all the paths.



$$\begin{aligned} \text{Path 1 : } t &\rightarrow x \rightarrow z \text{ gives } \frac{\partial z}{\partial x} \frac{dx}{dt} \\ \text{Path 2 : } t &\rightarrow y \rightarrow z \text{ gives } \frac{\partial z}{\partial y} \frac{dy}{dt} \end{aligned}$$

Worked Example 1

Let

$$z = x^2 y + e^y, \quad x = t^2, \quad y = \sin t.$$

Find $\frac{dz}{dt}$.

Solution

First compute the partial derivatives:

$$z_x = 2xy, \quad z_y = x^2 + e^y.$$

Also,

$$\frac{dx}{dt} = 2t, \quad \frac{dy}{dt} = \cos t.$$

Now apply the chain rule:

$$\frac{dz}{dt} = z_x \frac{dx}{dt} + z_y \frac{dy}{dt}.$$

So

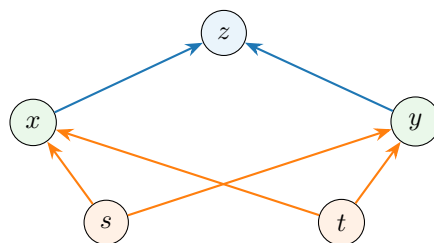
$$\frac{dz}{dt} = (2xy)(2t) + (x^2 + e^y) \cos t.$$

Finally substitute $x = t^2$, $y = \sin t$:

$$\frac{dz}{dt} = 4t^3 \sin t + (t^4 + e^{\sin t}) \cos t.$$

3 Two-variable chain rule: $z = f(x, y)$, $x = x(s, t)$, $y = y(s, t)$

Now x and y depend on **two independent variables** s and t . Therefore, we need **partial derivatives**.



Formulas

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

Worked Example 2

Let

$$z = x^2 + y^2, \quad x = u + v, \quad y = u - v.$$

Find z_u and z_v .

Solution

First,

$$z_x = 2x, \quad z_y = 2y.$$

Also,

$$x_u = 1, \quad y_u = 1, \quad x_v = 1, \quad y_v = -1.$$

Therefore,

$$z_u = z_x x_u + z_y y_u = 2x(1) + 2y(1) = 2x + 2y.$$

Substitute $x = u + v$, $y = u - v$:

$$z_u = 2(u + v) + 2(u - v) = 4u.$$

Similarly,

$$z_v = z_x x_v + z_y y_v = 2x(1) + 2y(-1) = 2x - 2y.$$

So

$$z_v = 2(u + v) - 2(u - v) = 4v.$$

Hence,

$$z_u = 4u, \quad z_v = 4v.$$

Quick check by direct substitution

Since

$$z = (u + v)^2 + (u - v)^2 = 2u^2 + 2v^2,$$

we immediately get

$$z_u = 4u, \quad z_v = 4v.$$

This matches the chain-rule answer.

4 Implicit differentiation: the key idea

Most important sentence in this handout

Implicit differentiation is not a new rule. It is simply the **chain rule** applied to an equation that remains equal to 0.

4.1 Case A: $F(x, y) = 0$, where $y = y(x)$

If y depends on x , then the equation really means

$$F(x, y(x)) = 0.$$

Differentiate both sides with respect to x :

$$\frac{d}{dx}(F(x, y(x))) = 0.$$

By the chain rule:

$$F_x \frac{dx}{dx} + F_y \frac{dy}{dx} = 0.$$

Since $\frac{dx}{dx} = 1$,

$$F_x + F_y \frac{dy}{dx} = 0.$$

Hence

$$\boxed{\frac{dy}{dx} = -\frac{F_x}{F_y}} \quad (F_y \neq 0).$$

Very common mistake

Do **not** reverse the numerator and denominator. The correct formula is

$$\boxed{\frac{dy}{dx} = -\frac{F_x}{F_y}}$$

not $-F_y/F_x$.

Worked Example 3: Circle

Let

$$F(x, y) = x^2 + y^2 - 25 = 0.$$

Then

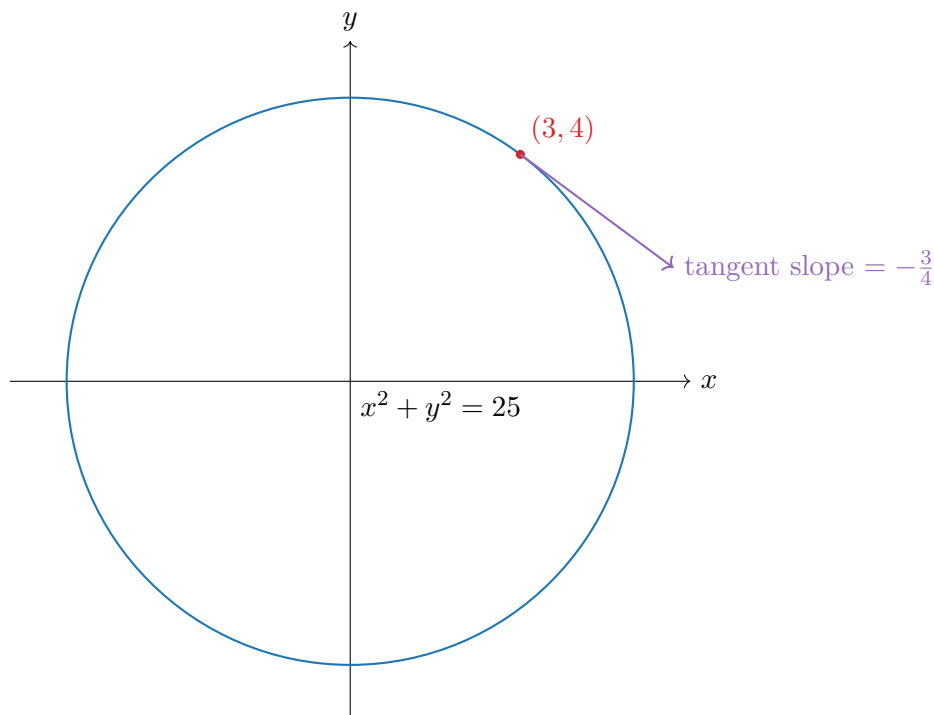
$$F_x = 2x, \quad F_y = 2y.$$

So

$$\frac{dy}{dx} = -\frac{2x}{2y} = -\frac{x}{y}.$$

At the point $(3, 4)$,

$$\boxed{\left. \frac{dy}{dx} \right|_{(3,4)} = -\frac{3}{4}}.$$



4.2 Case B: $F(x, y, z) = 0$, where $z = z(x, y)$

Now z is the dependent variable. So the equation really means

$$F(x, y, z(x, y)) = 0.$$

Differentiate partially with respect to x (keeping y constant):

$$F_x + F_z z_x = 0.$$

Hence

$$\boxed{z_x = -\frac{F_x}{F_z}} \quad (F_z \neq 0).$$

Similarly, differentiate with respect to y :

$$F_y + F_z z_y = 0,$$

so

$$\boxed{z_y = -\frac{F_y}{F_z}}.$$

Pattern

If an equation $F = 0$ hides one variable as a function of the others, then:

$$\text{derivative of dependent variable} = -\frac{\text{partial derivative wrt independent variable}}{\text{partial derivative wrt dependent variable}}.$$

Examples:

$$y_x = -\frac{F_x}{F_y}, \quad z_x = -\frac{F_x}{F_z}, \quad z_y = -\frac{F_y}{F_z}.$$

4.3 Case C: If $x = x(t)$, $y = y(t)$, $z = z(t)$ and $F(x, y, z) = 0$

This is the best place to see why implicit differentiation is really chain rule.

Since

$$F(x(t), y(t), z(t)) = 0,$$

differentiate with respect to t :

$$\frac{dF}{dt} = 0.$$

By the chain rule,

$$F_x \frac{dx}{dt} + F_y \frac{dy}{dt} + F_z \frac{dz}{dt} = 0.$$

So

$$\frac{dz}{dt} = -\frac{F_x \frac{dx}{dt} + F_y \frac{dy}{dt}}{F_z}.$$

Worked Example 4

Suppose a point moves on the surface

$$x^2 + y^2 + z^2 = 9$$

so

$$F(x, y, z) = x^2 + y^2 + z^2 - 9 = 0.$$

Let

$$x = t, \quad y = 2t.$$

Find $\frac{dz}{dt}$.

We have

$$F_x = 2x, \quad F_y = 2y, \quad F_z = 2z.$$

Also

$$\frac{dx}{dt} = 1, \quad \frac{dy}{dt} = 2.$$

Hence

$$\frac{dz}{dt} = -\frac{(2x)(1) + (2y)(2)}{2z} = -\frac{2x + 4y}{2z} = -\frac{x + 2y}{z}.$$

Now substitute $x = t$, $y = 2t$:

$$\frac{dz}{dt} = -\frac{5t}{z}.$$

5 A comparison table

Situation	What depends on what?	Main formula
Chain rule with one parameter	$z = f(x, y)$, $x = x(t)$, $y = y(t)$	$\frac{dz}{dt} = z_x \frac{dx}{dt} + z_y \frac{dy}{dt}$
Chain rule with two variables	$z = f(x, y)$, $x = x(s, t)$, $y = y(s, t)$	$z_s = z_x x_s + z_y y_s$, $z_t = z_x x_t + z_y y_t$
Implicit differentiation in two variables	$F(x, y) = 0$, $y = y(x)$	$\frac{dy}{dx} = -\frac{F_x}{F_y}$
Implicit differentiation in three variables	$F(x, y, z) = 0$, $z = z(x, y)$	$z_x = -\frac{F_x}{F_z}$, $z_y = -\frac{F_y}{F_z}$
Constrained motion in time	$F(x(t), y(t), z(t)) = 0$	$F_x x' + F_y y' + F_z z' = 0$

6 Common mistakes to avoid

Mistake checklist

1. Reversing $\frac{F_x}{F_y}$ and $\frac{F_y}{F_x}$ in $\frac{dy}{dx}$.
2. Forgetting that for chain rule you **add all the paths**.
3. Using ordinary derivatives instead of partial derivatives when there is more than one independent variable.
4. Forgetting which variable is the **dependent** variable in implicit differentiation.
5. Substituting numerical values too early, before forming the formula.

7 Mini practice

Try these on your own first.

1. If $z = x^2 + y^2$, $x = \cos t$, $y = \sin t$, find $\frac{dz}{dt}$.
2. If $z = xy$, $x = s + t$, $y = s - t$, find z_s and z_t .
3. For the curve $x^2 + xy + y^2 = 7$, find $\frac{dy}{dx}$.
4. If $F(x, y, z) = x^2 + y^2 + z^2 - 16 = 0$, find z_x and z_y .

Answers

1. $\frac{dz}{dt} = 0$, since $z = \cos^2 t + \sin^2 t = 1$.
2. $z = (s + t)(s - t) = s^2 - t^2$, so $z_s = 2s$, $z_t = -2t$.
3. Let $F(x, y) = x^2 + xy + y^2 - 7$. Then $F_x = 2x + y$, $F_y = x + 2y$, so

$$\frac{dy}{dx} = -\frac{2x + y}{x + 2y}.$$

4. Here $F_x = 2x$, $F_y = 2y$, $F_z = 2z$. Hence

$$z_x = -\frac{x}{z}, \quad z_y = -\frac{y}{z}.$$

Final takeaway

Whenever you feel confused, return to these two questions:

1. **Who depends on whom?**
2. **What are all the possible paths from the independent variable to the quantity I want to differentiate?**

If the problem is implicit, remember:

Differentiate the entire equation, and set the derivative equal to 0.

End of handout